

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and Infrastructure Engineering) SEMESTER - 5 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and Infrastructure Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCIC505 - ARTIFICIAL INTELLIGENCE

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 AI primarily focuses on developing systems that can:
 - A) Store large amounts of data
 - B) Perform fixed programmed tasks
 - C) Exhibit intelligent behaviour
 - D) Reduce computational cost
 - 2 Which is an example of an uninformed search?
 - A) A* Search
 - B) Greedy Best-First Search
 - C) BFS
 - D) Hill-Climbing
 - 3 Modelling a problem as a search problem requires defining:
 - A) Only the goal state
 - B) Only the initial state
 - C) State space + actions + goal + path cost
 - D) Only the heuristic
 - 4 Which of the following is used in Genetic Algorithms?
 - A) Minimax
 - B) Mutation & crossover
 - C) Constraint propagation
 - D) Backtracking
 - 5 Adversarial search is commonly used in:
 - A) Optimization
 - B) Game playing
 - C) Natural language processing
 - D) Image classification
 - 6 The Minimax algorithm assumes opponents are:
 - A) Random
 - B) Always cooperative
 - C) Always optimal
 - D) Unpredictable

- 7 Which operator is not part of propositional logic?
 - A) AND
 - B) OR
 - C) IMPLIES
 - D) LOOP
- 8 A formula is satisfiable if:
 - A) It is always false
 - B) It has at least one truth assignment that makes it true
 - C) It contains contradictions
 - D) It has no variables
- 9 In Bayesian network learning, parameters represent:
 - A) Costs
 - B) Conditional probability distributions
 - C) Logical rules
 - D) Rewards only
- 10 An MDP includes which of the following?
 - A) States, Rewards, Actions, Transition probabilities
 - B) States only
 - C) Rewards only
 - D) Deterministic paths only
- 11 Reinforcement learning involves learning through:
 - A) Labels
 - B) Rewards and punishments
 - C) Prior knowledge
 - D) Logical inference
- 12 Deep learning models mostly rely on:
 - A) Decision trees
 - B) Neural networks
 - C) Logic rules
 - D) Genetic operators

Q2. Solve the following.

- A) Define Artificial Intelligence and explain its scope in modern technology. 6
- B) What is a heuristic? Discuss the role of heuristic functions in problem-solving. 6

Q3. Solve the following.

- A) Explain the working principle of local search algorithms with a suitable example. 6
- B) What is a Constraint Satisfaction Problem (CSP)? Give an example. 6

Q4. Solve Any Two of the following.

- A) Explain propositional logic and its importance in AI. 6
- B) What is Alpha-Beta Pruning, and how does it improve Minimax? 6
- C) What is a Constraint Satisfaction Problem (CSP) in AI? 6

Q5. Solve Any Two of the following.

- A) Explain Bayesian learning and how it is applied in AI. 6
- B) Why is uncertainty important in AI, and how is it handled? 6
- C) What is a Markov Decision Process (MDP)? Explain with an example. 6

Q6. Solve Any Two of the following.

- A) Explain reinforcement learning and how an agent learns in an RL environment. 6
- B) Explain the difference between Model-Free and Model-Based Reinforcement Learning. 6
- C) What is Deep Reinforcement Learning and how does it differ from classical RL? 6

***** End *****